1.05 Images

Images can be stored in a computer system for the eventual purpose of displaying the image on a screen or for presenting it on paper, usually as a component of a document. Such an image can be created by using an appropriate graphics package. Alternatively, when an image already exists independently of the computer system, the image can be captured by using photography or by scanning.

Vector graphics

In an image that is created by a drawing package or a computer-aided design (CAD) package each component is an individual **drawing object**. The image is then stored, usually as a **vector graphic** file.

We do not need to consider how an image of this type would be created. We do need to consider how the data is stored after the image has been created. A vector graphic file contains a **drawing list**. The list contains a command for each object included in the image. Each command has a list of attributes, each attribute defines a **property** of the object. The properties include the basic geometric data such as, for a circle, the position of the centre and its radius. In addition, properties are defined such as the thickness and style of a line, the colour of a line and the colour that fills the shape. An example of what could be created as a vector graphic file is shown in Figure 1.05.

KEY TERMS

Vector graphic: a graphic consisting of drawing objects defined in a drawing list

Drawing object: a component defined by geometric formulae and associated properties

Drawing list: contains one set of values for each drawing object

Property: defines one aspect of the appearance of the drawing object

TASK 1.05

Construct a partial drawing list for the graphic shown in Figure 1.05. You can take measurements from the image and use the bottom left corner of the box as the origin of a coordinate system. You can invent your own format for the drawing list.

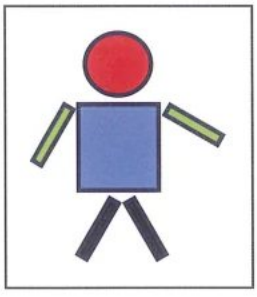


Figure 1.05 A simple example of a vector graphic image

The most important property of a vector graphic image is that the dimensions of the objects are not defined explicitly but instead are defined relative to an imaginary drawing canvas. In other words, the image is scalable. Whenever the image is to be displayed the file is read, the appropriate calculations are made and the objects are drawn to a suitable scale. If the user then requests that the image is redrawn at a larger scale the file is read again and another set

of calculations are made before the image is displayed. This avoids image distortion, such as the image appearing squashed or stretched.

Note that a vector graphic file can only be displayed directly on a graph plotter, which is an expensive specialised piece of hardware. For the image to appear correctly on other types of display, the vector graphic file often has to be converted to a bitmap.

Bitmaps

Most images do not consist of geometrically defined shapes, so a vector graphic representation is inappropriate. Instead, generally an image is stored as a bitmap. Typical uses are when capturing an existing image by scanning or perhaps by taking a screen-shot. Alternatively, an image can be created by using a simple graphics package.

The fundamental concept underlying the creation of a bitmap file is that the **picture element (pixel)** is the smallest identifiable component of a bitmap image. The image is stored as a two-dimensional matrix of pixels. The pixel itself is a very simple construct; it has a position in the matrix and it has a colour. It does not matter whether each pixel is a small rectangle, a small circle or a dot.

The scheme used to represent the colour has to be defined. The simplest option is to use one bit to represent the colour, so that the pixel is either black or white. Storage of the colour in four bits allows simple greyscale colouring. At least eight bits per pixel are necessary to provide a sufficient range of colours to provide a reasonably realistic representation of any image. The number of bits per pixel is sometimes referred to as the **colour depth**.

KEY TERMS

Picture element (pixel): the smallest identifiable component of a bitmap image, defined by just two properties: its position in the bitmap matrix and its colour

Colour depth: the number of bits used to represent one pixel

An alternative definition is the **bit depth**. Although these terms are sometimes used interchangeably, bit depth is best defined as the number of bits used to store each of the red, green and blue primary colours in the RGB colour scheme.

A colour depth of 8 bits per pixel provides 256 different colours. A bit depth of 8 bits per primary colour provides $256 \times 256 \times 256 = 16\,777\,216$ different colours. The eye cannot distinguish this number of different colours. However, this many are needed if an image contains areas of gradually changing colour such as in a picture of the sky. If a lower bit depth is used the image will show bands of colour.

We also need to decide which resolution to use for the image, which can be represented as the product of the number of pixels per row times the number of rows. When considering resolution it is important to distinguish between an **image resolution**, as defined in a bitmap file, and a **screen resolution** for a particular monitor screen that might be used to display the image. Both of these have to be considered if a screen display is being designed.

KEY TERMS

Bit depth: the number of bits used to represent each of the red, green and blue colours

Image resolution: the number of pixels in the bitmap file defined as the product of the width and the height values

Screen resolution: the product of width and height values for the number of pixels that the screen can display

A bitmap file does not define the physical size of a pixel or of the whole image. When the image is scaled the number of pixels in it does not change. If a well-designed image is presented on a suitable screen the human eye cannot distinguish the individual pixels. However, if the image is magnified too far the individual pixels will be seen. This is illustrated in Figure 1.06 which shows an original small image, a magnified version of this small image and a larger image created with a more sensible, higher resolution.



a b c

Figure 1.06 (a) a bitmap logo; (b) an over-magnified version of the image; (c) a sensible larger version

File size is always an issue with an image file. A large file occupies more memory space and takes longer to display or to be transmitted across a network. Usually, a vector graphic file uses considerably less memory space than a corresponding bitmap file.

You can calculate the size of a bitmap graphic knowing the resolution and the colour depth. As an example, consider that a bitmap graphic is needed to fill a laptop screen where the resolution is 1366 by 768. If we want colour depth of 24 then the number of bits we need is:

$$1366 \times 768 \times 24 = 25\,178\,112 \text{ bits}$$

The result of this calculation shows the number of bits, but a size is always quoted as a number of bytes or multiples of bytes. For our bitmap graphic:

25 178 112 bits = 25 178 112 ÷ 8 = 3 147 264 bytes

= 3 147 264 ÷ 1024 = 3073.5 kibibytes (3073.5 KiB)

= 3073.5 ÷ 1024 = approximately 3 MiB

Note that this calculation has assumed that the colour depth specifies the total number of bits used to define each pixel. If the information given was that the bit depth was eight, then the calculation would use 8 + 8 + 8 for the number of bits per pixel.

WORKED EXAMPLE 1.04

You have been asked to calculate a value for the minimum size of a bitmap file. The bitmap is to use a bit depth of 8 and the bitmap is to be printed with 72 dpi (dots per inch) and to have dimensions 5 inches by 3 inches.

We use the information provided about the colour depth or the bit depth to give the number of bits per pixel. In this case the bit depth is 8, which means 8 bits for each of the RGB components, so 24 bits are needed for one pixel.

Let's state that 72 dpi means 72 pixels per inch.

(Continued)

So, the number of pixels per row is 5 × 72 = 360

And the number of pixels per column is 3 × 72 = 216

Therefore, the total number of pixels is 360 × 216 = 77 760

The total number of bits is this value multiplied by 24. However, we want the size in bytes not bits, so we multiply by 3 because there are 8 bits in a byte. So, we get:

77 760 × 3 = 233 280 bytes.

We can quote this in kibibytes by dividing by 1024:

233 280 / 1024 = 227.8 KiB

A bitmap file has to store the pixel data that defines the graphic, but the file must also have a **file header** that contains information on how the graphic has been constructed. Because of this, the bitmap file size is larger than the size of the graphic alone. At the very least the header will define the colour depth or bit depth and the resolution.



KEY TERM

File header: a set of bytes at the beginning of a bitmap file which identifies the file and contains information about the coding used

The following are considerations when justifying the use of either a bit map or a vector graphic for a specific task.

- A vector graphic is chosen if a diagram is needed to be constructed for part of an architectural, engineering or manufacturing design.
- If a vector graphic file has been created but there is a need to print a copy using a laser or inkjet printer the file has first to be converted to a bitmap.
- A digital camera automatically produces a bitmap.
- A bitmap file is the choice for insertion of an image into a document, publication or web page.

1.06 Sound

Natural sound consists of variations in pressure which are detected by the human ear. A typical sound contains a large number of individual waves, each with a defined frequency. The result is a wave form in which the amplitude of the sound varies in a continuous but irregular pattern.

If we want to store sound or transmit it electronically the original analogue sound signal has to be converted to a binary code. The measured sound values are input to a sound encoder which has two components. The first is a band-limiting filter. This is needed to remove high-frequency components. A human ear cannot detect these very high frequencies and they could cause problems for the coding if not removed. The other component in the encoder is an analogue-to-digital converter (ADC) which converts the **analogue data** to **digital data**.

Figure 1.07 shows the **sampling** operation of the ADC. The amplitude of the wave (the red line) has to be sampled at regular intervals. The blue vertical lines indicate the sampling times. The amplitude cannot be measured exactly; instead the amplitude is approximated

by the closest of the defined amplitudes represented by the horizontal lines. In Figure 1.07, sample values 1 and 4 will be an accurate estimate of the actual amplitude because the wave is touching an amplitude line. In contrast, samples 5 and 6 will not be accurate because the actual amplitude is approximately half way between the two closest defined values.

To code sound, we need to make two decisions. The first is the number of bits we will use to store the amplitude values, which defines the **sampling resolution**. If we use only three bits then eight levels can be defined as shown in Figure 1.07. If too few are used there will be a significant error when the closest amplitude in the scale of values dictated by the sampling resolution is used as the approximation for the real value. In practice, 16 bits provides reasonable accuracy for most digitised sound.

We also need to choose the **sampling rate**, which is the number of samples taken per second. This should be in accordance with Nyquist's theorem which states that sampling must be done at a frequency at least twice the highest frequency of the sound in the sample.

KEY TERMS

Analogue data: data obtained by measurement of a physical property which can have any value from a continuous range of values

Digital data: data that has been stored as a binary value which can have one of a discrete range of values

Sampling: taking measurements at regular intervals and storing the value

Sampling resolution: the number of bits used to store each sample

Sampling rate: the number of samples taken per second

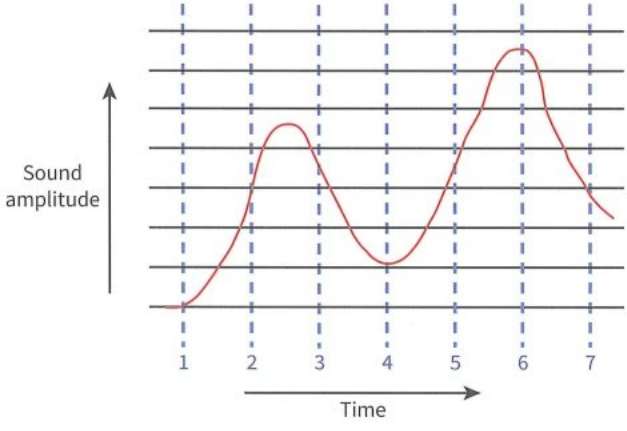


Figure 1.07 ADC sampling

Once again file size can be an issue. An increased sampling rate and an increased sampling resolution will both cause an increase in file size.

1.07 Compression techniques

Larger files require larger storage capacity but more importantly, larger files have lower transmission or download rates. For this reason, compression techniques are often used to reduce file size.

There are two categories of compression. The first is **lossless compression** where the file size is reduced but no information is lost. The process can be reversed to re-create the